BRAC University
MAT-215
Exercise Sheet # 5

1. Expand each of the following functions in a Taylor series about the indicated points:

$(i)\, e^{-z}$ at $z = 0$

$(ii)\, \cos z$ at $z = \frac{\pi}{2}$

$(iii)\, z^3 - z^2 + 4z - 2$ at $z = 2$

$(iv)\, z e^{2z}$ at $z = -1$.

2. Expand $f(z) = \dfrac{z}{(z-1)(2-z)}$ in a

Laurent series valid for

$(i)\, |z| < 1$ $(ii)\, 1 < |z| < 2$

$(iii)\, |z| > 2$ $(iv)\, |z-1| > 1$

$(v)\, 0 < |z-2| < 1$.

3. Expand $f(z) = \dfrac{1}{z(z-2)}$ in a

Laurent series valid for

$(i)\, 0 < |z| < 2$

$(ii)\, |z| > 2$

4. Evaluate $\oint_C \dfrac{z^2}{2z^2 + 5z + 2}\, dz$ using the residue at the poles, where C is the unit circle

$|z| = 1$.

5. Evaluate $\oint_C \dfrac{z^2 + 4}{z^3 + 2z^2 + 2z}\, dz$ using the residue at the poles, around the circle $|z| = 3$.

6. Evaluate $\oint_C \dfrac{z e^{i\pi z}}{(z^2 + 2z + 5)(z^2 + 1)^2}\, dz$ using the residue at the poles, where C is the upper

half circle of the equation $|z| = 2$.

7. Evaluate $\dfrac{1}{2\pi i} \oint_C \dfrac{z^2 - z + 2}{z^4 + 10z^2 + 9}\, dz$ using the residue at the poles, around the circle C with

the equation $|z| = 4$.

8. For each of the following functions locate and name the singularities in the finite z-plane and determine whether they are isolated singularities or not.

(a) $f(z) = \dfrac{z}{(z^2+4)^2}$; [Hint: f(z) = $\dfrac{z}{\{z^2-(2i)^2\}^2}$]

(b) $f(z) = \dfrac{\ln(z-2)}{(z^2+2z+4)^4}$